

FITBUDDY - AI FITNESS PLAN GENERATOR USING GEMINI MODELS

Phase 2: Requirement Analysis Phase

This phase focuses on identifying the functional and technical requirements needed to develop FitBuddy.

Functional Requirements

The system should be able to:

- Collect user information such as:
 - Name
 - User ID
 - Age
 - Weight
 - Fitness goal
 - Workout intensity
- Generate personalized 7-day workout plans.
- Generate nutrition or recovery tips.
- Accept user feedback.
- Modify existing workout plans based on feedback.
- Store user information and workout plans.
- Display original and updated workout plans.
- Provide an administrative view of registered users.

Technical Requirements

The application requires:

- Python
- FastAPI
- Uvicorn
- Google Gemini API
- SQLAlchemy
- SQLite
- Jinja2
- HTML and CSS
- Git
- Virtual environment

AI Requirements

The AI system should:

- Understand user fitness goals.
- Generate structured workout plans.
- Produce concise nutrition/recovery recommendations.
- Modify plans according to user feedback.
- Provide responses suitable for real-time web interaction.

Outcome:

A documented set of functional, technical, and AI requirements.